

WAINWRIGHT, AK, USA

WMO: 700300

Lat: 70.639N

Lon: 159.995W

Elev: 30

StdP: 14.68

Time Zone: -9.00 (NAK)

Period: 99-19

WBAN: 27503

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -37.6	(c) -34.1	(d) -35.1	(e) 0.7	(f) -28.2	(g) -33.7	(h) 0.8	(i) -27.5	(j) 36.3	(k) 3.1	(l) 33.3	(m) 2.7	(n) 5.5	(o) 140	(p) 1.063

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.9	63.1	54.9	59.0	52.9	55.2	50.7	56.0	61.6	53.4	58.0	51.0	55.3	11.8	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
53.0	59.9	58.9	50.3	54.0	56.2	47.8	49.3	53.6	23.8	61.4	22.2	58.4	20.8	55.5	65.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 31.3	(b) 27.4	(c) 24.9	(e) -40.1	(f) 71.7	(g) 6.4	(h) 5.3	(i) -44.7	(j) 75.5	(k) -48.4	(l) 78.6	(m) -52.0	(n) 81.6	(o) -56.6	(p) 85.4		
(4) 31.3	(5) 27.4	(6) 24.9	(7) -40.1	(8) 71.7	(9) 6.4	(10) 5.3	(11) -44.7	(12) 75.5	(13) -48.4	(14) 78.6	(15) -52.0	(16) 81.6	(17) -56.6	(18) 85.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.5	-10.6	-9.0	-10.4	3.2	24.2	39.2	45.6	43.0	35.9	22.5	5.9	-4.9		
	DBStd	23.62	13.44	15.58	13.24	11.14	9.58	6.85	7.06	6.53	5.98	7.84	11.58	11.86		
	HDD50	12649	1877	1651	1873	1403	801	333	173	235	424	853	1324	1702		
	HDD65	18058	2342	2071	2338	1853	1266	775	600	681	873	1318	1774	2167		
	CDD50	68	0	0	0	0	0	9	38	20	1	0	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	4	0	0	0	0	0	0	3	1	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	12.1	11.5	12.8	12.0	12.3	12.4	11.8	11.7	11.4	11.5	12.7	12.6	12.1		
	PrecAvg	9.9	0.5	0.5	0.4	0.3	0.4	0.5	1.3	2.1	1.4	1.3	0.6	0.5		
(15) Precipitation	PrecMax	20.1	1.5	1.4	3.1	1.0	1.3	1.8	2.8	4.5	5.0	2.9	2.8	1.8		
	PrecMin	5.0	0.0	0.1	0.0	0.0	0.0	0.0	0.4	0.7	0.3	0.3	0.2	0.1		
	PrecStd	3.4	0.4	0.4	0.6	0.2	0.3	0.4	0.6	1.0	1.0	0.6	0.5	0.4		
	PrecStd	3.4	0.4	0.4	0.6	0.2	0.3	0.4	0.6	1.0	1.0	0.6	0.5	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.3	33.4	35.1	36.0	48.3	66.2	71.4	65.5	55.4	39.5	33.4	29.7		
		MCWB	29.7	30.0	29.7	34.2	43.2	55.6	59.4	57.5	49.9	38.2	32.1	28.6		
	2%	DB	25.2	27.8	25.2	29.9	41.6	59.5	64.4	60.6	51.6	36.5	31.0	25.2		
		MCWB	24.3	26.2	23.5	28.6	38.6	52.3	55.5	54.2	47.8	35.6	29.6	24.1		
	5%	DB	16.0	21.2	18.2	25.2	37.5	54.3	61.0	56.0	48.1	33.8	27.5	18.2		
		MCWB	15.4	20.3	17.5	24.0	35.6	48.8	53.7	52.1	45.8	32.2	26.3	17.6		
	10%	DB	7.5	14.3	8.1	19.7	35.3	49.8	56.9	53.2	45.2	31.7	22.9	11.9		
		MCWB	6.6	13.4	6.9	19.0	33.7	45.7	51.6	50.4	43.6	30.4	21.8	11.2		
	0.4%	WB	30.0	31.2	31.7	34.2	43.5	56.4	60.2	58.5	50.7	38.8	32.5	29.0		
		MCDB	30.3	33.2	35.4	35.5	48.5	64.1	69.1	62.8	54.8	39.4	33.0	29.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	24.9	26.8	23.9	28.6	38.6	52.5	56.9	55.1	48.1	35.8	29.8	24.4		
		MCDB	25.8	28.3	25.8	29.6	41.4	58.8	63.6	58.6	51.1	36.5	30.8	25.4		
	5%	WB	16.6	21.1	17.7	24.2	35.7	49.0	54.3	52.6	46.1	32.4	26.3	17.5		
		MCDB	17.3	22.2	18.6	25.4	37.5	54.0	59.7	55.7	48.1	33.2	27.6	18.0		
	10%	WB	8.0	14.9	7.9	19.0	33.8	46.0	51.8	50.1	43.7	30.5	21.8	11.5		
		MCDB	8.9	15.8	8.8	19.6	35.1	49.6	56.7	53.2	45.4	31.3	22.7	12.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.4	12.8	12.3	12.8	8.5	11.7	12.9	10.0	7.4	8.0	11.4	12.0		
		MCDBR	19.6	19.4	16.7	14.6	10.5	20.5	20.3	16.1	12.3	7.3	10.7	15.9		
		MCWBR	19.6	18.6	15.6	14.2	8.3	14.0	12.7	10.7	8.7	7.0	10.4	15.7		
	5% WB	MCDBR	19.5	19.3	16.4	14.7	10.4	19.9	19.1	14.5	11.7	7.5	11.1	16.2		
		MCWBR	19.6	18.6	15.4	14.2	8.1	14.0	12.6	10.1	8.5	7.3	10.8	16.1		
			(d) N/A	(e) 0.205	(f) 0.240	(g) 0.304	(h) 0.327	(i) 0.324	(j) 0.337	(k) 0.322	(l) 0.290	(m) 0.204	(n) N/A	(o) N/A		
(40) Clear-Sky Solar Irradiance			(d) N/A	(e) 2.019	(f) 2.217	(g) 2.091	(h) 2.164	(i) 2.406	(j) 2.473	(k) 2.558	(l) 2.547	(m) 2.197	(n) N/A	(o) N/A		
			(d) 0	(e) 161	(f) 247	(g) 258	(h) 267	(i) 275	(j) 266	(k) 258	(l) 234	(m) 166	(n) 0	(o) 0		
			(d) 0	(e) 12	(f) 21	(g) 35	(h) 37	(i) 30	(j) 28	(k) 23	(l) 16	(m) 9	(n) 0	(o) 0		
			(d) 0	(e) 99	(f) 597	(g) 1313	(h) 1726	(i) 1866	(j) 1514	(k) 914	(l) 451	(m) 120	(n) 0	(o) 0		
(44) All-Sky Solar Radiation			(d) 0	(e) 10	(f) 34	(g) 87	(h) 165	(i) 132	(j) 168	(k) 89	(l) 39	(m) 15	(n) 0	(o) 0		
			(d) 0	(e) 10	(f) 34	(g) 87	(h) 165	(i) 132	(j) 168	(k) 89	(l) 39	(m) 15	(n) 0	(o) 0		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (4 neighbors)		+2.63	+4.82	+2.48	N/A	N/A	N/A	-869	-873	N/A	N/A	

Nomenclature: See separate page